

Count the Overlap Once

Topic 4.4 · TODAY'S PROBLEM

Suppose 120 students answer two yes-or-no questions: “Do you like pizza?” and “Do you like tacos?” The table summarizes their answers. Let P be the event that a randomly selected student likes pizza and T the event that the student likes tacos.

Jamie: “ $P(P \text{ or } T) = 80/120 + 60/120 = 140/120$. Because a probability cannot exceed 1, this must mean everyone likes at least one; I would report the answer as 1.”

Noor: “Some students may have been counted in both totals, so we need the intersection before deciding.”

Student food preferences (counts)

Preference	Pizza	Not pizza	Total
Tacos	35	25	60
Not tacos	45	15	60
Total	80	40	120

FIRST TAKE (5 min, on your sheet)

Does the table support reporting everyone likes pizza or tacos? Choose a claim and cite one count.

- DOK 1** (a) State the number of students in $P \cap T$ and write $P(P \cap T)$ as a fraction.
- DOK 2** (b) Compute $P(P \cup T)$ so every student in the union is counted once. Also compute the probability that a student likes neither food.
- DOK 3** (c) Decide whether P and T are mutually exclusive and which student’s reasoning is warranted. Cite the 35-student intersection and your union probability, explain what Jamie’s calculation would mislead a reader into believing, and name a pair of events from this context that really are mutually exclusive.

Video + follow-along: u4_lesson3-4-5_live.html (scan the code)

Finish (a)–(c) after the video. Turn in your sheet.

